

Architecture of Massively Scalable Applications, Spring 2026
Lab 9 Installation Guide

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Setting up Minikube for your local Kubernetes cluster

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Before You Begin

This guide installs **Minikube V1.38.1** and a compatible **kubectl** on your machine, then starts a local single-node Kubernetes cluster using Docker as the driver. By the end, you will have a working cluster you can stop, restart, and redeploy workloads to throughout the rest of the course.

What gets installed

- **Minikube V1.38.1** — the tool that runs a local Kubernetes cluster inside a Docker container.

What you already have. Docker Desktop (from Lab 3) ships with **kubectl** built in and adds it to your PATH automatically. You do *not* need to install kubectl separately — plain **kubectl** commands will work as soon as Docker Desktop is installed. As a fallback, Minikube also bundles its own copy you can invoke as **minikube kubectl -- <cmd>**.

Prerequisites

- **Docker Desktop installed and running** from Lab 3. Minikube uses Docker as its driver, so Docker must be up before you start the cluster. Look for the whale icon in your system tray / menu bar — it must be steady (not animating).
- **At least 8 GB total system RAM.** Minikube needs 4 GB minimum for the cluster; 8 GB works, 16 GB is comfortable.
- **At least 20 GB free disk space** for Minikube’s VM, Kubernetes images, and your workloads. Windows users short on C: drive space can follow the alternative Part A installation on the D: drive.
- **Hardware virtualization enabled** (Intel VT-x or AMD-V) in BIOS. If Docker Desktop already works for you, this is already true.

Why we pin to V1.38.1 Minikube is a fast-moving project. Pinning the whole class to V1.38.1 means every student, TA, and grader runs the same CLI behavior. If you install “latest”, a new release mid-semester can change flags, defaults, or error messages in ways that break these instructions. Use V1.38.1 unless a course announcement says otherwise.

Administrator / sudo access is required.

- **Windows:** every command in Part A must be run in a *PowerShell as Administrator* window. Without admin rights, you cannot modify the system Path, and the setup will silently fail.
- **macOS / Linux:** a few commands use **sudo** to move binaries into **/usr/local/bin**. Have your password ready.

One warning about timing. The first run of `minikube start` downloads the Kubernetes base image (roughly 300–500 MB). This can take **3 to 5 minutes** on a good connection, and noticeably longer on a slow one. The terminal may appear frozen at various points — it is not. **Do not cancel it.** If you cancel mid-pull, the next start has to redo the work.

Important: always control the cluster from the terminal, never from the Docker Desktop GUI. Two specific traps to avoid:

- **Leave Docker Desktop’s “Enable Kubernetes” toggle OFF.** Docker Desktop has its own Kubernetes option under *Settings* → *Kubernetes*. If you turn it on, you start a *different* cluster that is not Minikube, and `kubectl` will target the wrong one. Never enable it for this course.
- **Do not press Stop on the Minikube container in Docker Desktop’s Containers tab.** When Minikube is running, you will see a container (usually named `minikube`) in Docker Desktop. Stopping it from the GUI is a hard kill and can corrupt cluster state. Always use `minikube stop` from the terminal.

Part A — Windows Installation (Primary: C: Drive)

This is the default path. Follow steps A1 through A6 in order. If you are running low on C: drive space, skip to the alternative Part A section that follows.

Step A1 — Open PowerShell as Administrator

Press the Start key, type PowerShell, right-click **Windows PowerShell**, and choose **Run as administrator**. Click **Yes** on the UAC prompt.

If the window title bar does **not** say “Administrator”, close it and repeat. The remaining steps fail silently without admin rights.

Step A2 — Create the installation folder

```
New-Item -ItemType Directory -Path "C:\minikube" -Force | Out-Null
```

Creates C:\minikube if it does not exist. The -Force flag makes it a no-op if the folder is already there, so this is safe to re-run.

Step A3 — Download Minikube V1.38.1

First, enforce TLS 1.2 for the download. Some Windows defaults are too old to negotiate with GitHub. Split into two statements so each paste is a complete PowerShell command:

```
$proto = [Net.SecurityProtocolType]::Tls12  
[Net.ServicePointManager]::SecurityProtocol = $proto
```

Next, set up the download URL. We split it across variables so no single line gets too long for the page:

```
$base = "https://github.com/kubernetes/minikube/releases/download"  
$rel  = "v1.38.1/minikube-windows-amd64.exe"  
$out  = "C:\minikube\minikube.exe"
```

Then download the binary into C:\minikube:

```
$ProgressPreference = "SilentlyContinue"  
Invoke-WebRequest -Uri "$base/$rel" -OutFile $out
```

Setting `$ProgressPreference` suppresses the progress bar, which otherwise slows the download significantly. The file is about 90 MB — expect **30 seconds to 2 minutes** depending on your connection. **Do not interrupt this step.**

Step A4 — Add C:\minikube to the system PATH

This makes `minikube` runnable by name from any terminal. We check the existing PATH first so we never add a duplicate entry:

```
$machinePath = [Environment]::GetEnvironmentVariable("Path", "Machine")
if ($machinePath -notlike "*C:\minikube*") {
    if ($machinePath.EndsWith(";")) {
        $newPath = $machinePath + "C:\minikube"
    } else {
        $newPath = $machinePath + ";C:\minikube"
    }
    [Environment]::SetEnvironmentVariable("Path", $newPath, "Machine")
}
$env:Path = [Environment]::GetEnvironmentVariable("Path", "Machine")
```

The last line refreshes the current PowerShell session's PATH so you do not have to reopen the terminal immediately. The persistent change applies to all future terminals.

Step A5 — Verify the installation

```
minikube version
```

You should see `minikube version: V1.38.1`. If you get “minikube is not recognized”, close **all** PowerShell windows, open a new one as administrator, and try again — PATH changes only take effect in new terminals.

Step A6 — Set Docker as the default driver

```
minikube config set driver docker
```

This records your choice so you don't have to pass `--driver=docker` on every start. Part D will start the cluster using this default.

Windows primary installation complete. Skip to Part D to start the cluster.

Part A (Alternative) — Windows on D: Drive

Use this path only if your C: drive is low on space. Minikube's image cache and VM data grow steadily over the semester; if you have less than roughly 15 GB free on C:, install to D: instead. The steps are identical to the primary Part A but with D: paths, plus one extra command to relocate Minikube's data directory.

Step 1 — Open PowerShell as Administrator

Same as primary Part A, Step A1.

Step 2 — Create both folders

```
New-Item -ItemType Directory -Path "D:\minikube" -Force | Out-Null
New-Item -ItemType Directory -Path "D:\.minikube" -Force | Out-Null
```

The first folder holds the `minikube.exe` binary. The second will hold Minikube's data — VM disk, cached images, logs — which is what grows over time.

Step 3 — Download Minikube V1.38.1 to D:

Enforce TLS 1.2 (two separate statements so each paste is complete):

```
$proto = [Net.SecurityProtocolType]::Tls12
[Net.ServicePointManager]::SecurityProtocol = $proto
```

Set up the download URL as variables:

```
$base = "https://github.com/kubernetes/minikube/releases/download"
$rel  = "v1.38.1/minikube-windows-amd64.exe"
$out  = "D:\minikube\minikube.exe"
```

Then download the binary into D:\minikube:

```
$ProgressPreference = "SilentlyContinue"
Invoke-WebRequest -Uri "$base/$rel" -OutFile $out
```

Step 4 — Add D:\minikube to the system PATH

```
$machinePath = [Environment]::GetEnvironmentVariable("Path", "Machine")
if ($machinePath -notlike "*D:\minikube*") {
    if ($machinePath.EndsWith(";")) {
        $newPath = $machinePath + "D:\minikube"
    } else {
        $newPath = $machinePath + ";D:\minikube"
    }
    [Environment]::SetEnvironmentVariable("Path", $newPath, "Machine")
}
$env:Path = [Environment]::GetEnvironmentVariable("Path", "Machine")
```

Step 5 — Point MINIKUBE_HOME to D:

This is the critical step for the alternative path. Without it, Minikube still writes its multi-gigabyte data directory to %USERPROFILE%\minikube on C:, which defeats the purpose of the D: install.

```
$mh = "D:\minikube"
[Environment]::SetEnvironmentVariable("MINIKUBE_HOME", $mh, "Machine")
$env:MINIKUBE_HOME = $mh
```

Step 6 — Verify and set Docker driver

```
minikube version
minikube config set driver docker
```

You should see `minikube version: V1.38.1`. The driver config is written under the new `MINIKUBE_HOME` you set in Step 5, inside `config/config.json`. Continue to Part D to start the cluster.

Part B — macOS Installation

Pick the download that matches your Mac's chip. If you are unsure, run `uname -m`: `x86_64` is Intel, `arm64` is Apple Silicon (M1/M2/M3/M4).

Step 1 — Open Terminal

Open **Terminal** from Applications → Utilities, or with Spotlight (Cmd+Space, then type “Terminal”).

Step 2 — Download Minikube V1.38.1

Set up the URL base once:

```
BASE="https://github.com/kubernetes/minikube/releases/download"  
VER="v1.38.1"
```

Intel Macs:

```
curl -LO "$BASE/$VER/minikube-darwin-amd64"
```

Apple Silicon Macs:

```
curl -LO "$BASE/$VER/minikube-darwin-arm64"
```

Expect 30 seconds to 2 minutes.

Step 3 — Install the binary to `/usr/local/bin`

Intel Macs:

```
sudo install minikube-darwin-amd64 /usr/local/bin/minikube  
rm minikube-darwin-amd64
```

Apple Silicon Macs:

```
sudo install minikube-darwin-arm64 /usr/local/bin/minikube  
rm minikube-darwin-arm64
```

`sudo install` copies the file, sets executable permissions, and places it on your PATH in one step.

Step 4 — Verify and set Docker driver

```
minikube version  
minikube config set driver docker
```

You should see `minikube version: V1.38.1`. Continue to Part D to start the cluster.

Part C — Linux Installation (Ubuntu / Debian / WSL2)

These instructions cover native Ubuntu/Debian and Ubuntu-on-WSL2. For other distributions, adapt the package manager commands.

Step 1 — Download Minikube V1.38.1

```
BASE="https://github.com/kubernetes/minikube/releases/download"
VER="v1.38.1"
curl -LO "$BASE/$VER/minikube-linux-amd64"
```

Step 2 — Install the binary to /usr/local/bin

```
sudo install minikube-linux-amd64 /usr/local/bin/minikube
rm minikube-linux-amd64
```

Step 3 — Verify and set Docker driver

```
minikube version
minikube config set driver docker
```

Optional: .deb package alternative. If you prefer package management over a raw binary:

```
BASE="https://github.com/kubernetes/minikube/releases/download"
VER="v1.38.1"
FILE="minikube_1.39.0-0_amd64.deb"
curl -LO "$BASE/$VER/$FILE"
sudo dpkg -i "$FILE"
rm "$FILE"
```

The end result is the same: a minikube binary on your PATH. Proceed to Part D.

Part D — Starting the Cluster

Step D1 — Confirm Docker Desktop is running

Open Docker Desktop (if not already open) and wait for the whale icon to stabilize — no animation, no “Docker Desktop starting” banner. Without this, `minikube start` fails within seconds with a “cannot connect to Docker daemon” error.

Step D2 — Pick a memory tier and save it as your default

Minikube reserves RAM from your host for the Kubernetes control plane plus any workloads you deploy. Pick a tier based on your total system memory.

Tier	Memory (MB)	Use if you have...
Minimum	4096	8 GB total RAM. Works for single-service labs; may feel tight later in the course.
Recommended	6144	16 GB total RAM. Balanced. This is what we use and test against.
Comfortable	8192	16+ GB RAM with few other apps open. Good headroom for multi-pod deployments.

Save your choice as the profile default:

```
minikube config set memory 6144
```

Replace 6144 with 4096 or 8192 if you picked a different tier. Combined with the driver config you set in Parts A/B/C, this means future `minikube start` commands need **no flags at all** — Minikube reads driver and memory from your saved config every time.

If your machine has only 8 GB total, use 4096 and close heavy apps (browsers, IntelliJ, extra Docker containers) before starting the cluster.

Step D3 — Start the cluster

This step takes 3 to 5 minutes the first time. Minikube is downloading the Kubernetes base image and setting up the control plane. The output may pause for 30 seconds or more between lines — that is normal. **Do not cancel with Ctrl+C.**

```
minikube start
```

No flags needed — Minikube uses the driver and memory you saved in config. When it finishes, you will see output ending in something like **Done! kubectl is now configured to use "minikube" cluster...**

Step D4 — Verify the cluster is running

Plain `kubectl` works here because Docker Desktop put it on your PATH:

```
kubectl version --client
```

You should see a `Client Version: v1.X.Y` line. Then check the node:

```
kubectl get nodes
```

You should see a single node with status `Ready`. If the status is `NotReady`, wait 20–30 seconds and try again — the node takes a moment after start to register.

```
minikube status
```

All lines should read `Running` or `Configured`. If any line says `Stopped`, the cluster did not fully start — see the troubleshooting appendix.

Step D5 — (Optional) Open the Kubernetes dashboard

```
minikube dashboard
```

This launches a browser tab with a web UI for inspecting pods, deployments, services, and logs. It is useful for quick visual inspection during labs. Close the terminal tab running this command to shut the dashboard down; the cluster itself keeps running.

Part E — Managing Memory and Cluster Lifecycle

Daily commands

```
minikube stop      # shut the cluster cleanly (data stays)
minikube start     # start it back up (uses saved config)
minikube status    # show node and control plane state
minikube delete    # destroy the cluster and all its data
```

`stop/start` is your daily workflow: stop when you're done for the day, start when you want to work again. Your deployments persist across stop/start cycles.

Changing the memory limit

There is **no in-place resize** for a running Minikube cluster. To change memory, you must delete and recreate it. Because your workloads are defined in YAML manifests that you can re-apply in seconds, this is a cheap operation.

```
minikube stop
minikube delete
minikube config set memory 8192
minikube start
```

The `minikube config set memory 8192` line updates your saved default. The fresh `minikube start` then picks that value up. After this, re-apply any deployments with `kubectl apply -f <your-yaml>` as usual.

Note: `minikube config set memory` on its own **does not** resize a running cluster — it only sets the default that the *next* `minikube start` will read. That's why the `delete` step above is required.

Preserving state across cluster recreation

For this course, the source of truth for everything running in your cluster is your YAML manifests, checked into git. After `minikube delete && minikube start`, a quick `kubectl apply -f .` brings everything back. If you ever create workloads that carry persistent state (databases using `PersistentVolumeClaim`), back their data up *before* deleting the cluster — `minikube delete` wipes volumes along with everything else.

Appendix — Troubleshooting

minikube is not recognized (Windows). PATH was not refreshed. Close **all** PowerShell windows, open a new one, and try again. If it still fails, confirm the folder is on PATH:

```
$env:Path -split ';' | Select-String minikube
```

If nothing prints, re-run the PATH command from Part A Step A4.

kubectl is not recognized. On all platforms, `kubectl` comes from Docker Desktop. If the command is not found, Docker Desktop is either not installed or its PATH entry is missing. Open Docker Desktop, confirm it runs, then close and reopen your terminal. As a fallback that always works, use `minikube kubectl -- <cmd>` to invoke the bundled `kubectl` directly.

“Cannot connect to the Docker daemon.” Docker Desktop is not running or still starting. Open Docker Desktop, wait for the whale icon to go steady, then retry `minikube start`.

minikube start appears frozen. Almost always, it is downloading an image. First starts pull 300–500 MB; subsequent starts are much faster. Give it at least 5 minutes before assuming something is wrong. If you have definitely lost network connectivity, cancel with Ctrl+C, restore the network, then run `minikube start` again — it resumes from where it left off.

“Requested memory XXXX MB is less than the recommended minimum.” You passed a value below Minikube’s safety floor (about 1800 MB). Use at least 4096.

“VT-x/AMD-V is not enabled” or similar virtualization error. Hardware virtualization is disabled in your BIOS/UEFI. You must enable it (look for “Intel VT-x”, “AMD-V”, or “Virtualization Technology” in BIOS settings). On Windows, also verify Hyper-V or WSL2 is enabled per Lab 3’s prerequisites.

Running out of disk space during start. If you installed to C: and are hitting space limits, switch to the alternative Part A (D: drive) path. You must first run `minikube delete`, then follow the D: steps — that relocates Minikube’s data directory via `MINIKUBE_HOME`.

kubectl version mismatch warnings. Standalone `kubectl` and Minikube’s bundled Kubernetes can drift by a minor version or two without real issues. If a specific command fails with a version-related error, fall back to the bundled client:

```
minikube kubectl -- <your command>
```

Cluster stuck in a bad state (standard reset). Works for most problems. You lose cluster state but keep your YAML manifests:

```
minikube delete
minikube start
```

Cluster still broken (nuclear reset). If the standard reset doesn't fix it, wipe every Minikube profile and the entire data directory:

```
minikube delete --all --purge
```

`--all` removes every Minikube profile on your machine (not just the default). `--purge` wipes the `MINIKUBE_HOME` directory itself — cached ISOs, images, logs, and config. After this your saved driver and memory defaults are gone; re-run `minikube config set driver docker` and `minikube config set memory 6144`, then `minikube start`.

References:

[*Official Minikube Start Guide*](#)

[*Minikube v1.38.1 Release Notes*](#)

[*Kubernetes kubectl Installation Docs*](#)